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Research Article

Modeling of Marshall Stability of plastic-reinforced asphalt concrete using machine learning algorithms and SHAP

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ABSTRACT

Pavement engineering has consistently emphasized the need to improve the performance and longevity of asphalt concrete, a critical material in road construction. This study focuses on the predicting the Marshall Stability of plastic-reinforced asphalt concrete using machine learning algorithms. Machine learning algorithms, including Random Forest (RF), Gradient Boosting, Extreme Gradient Boosting (XGB), and Bagging Regressor (BR) were employed to predict the Marshall Stability. The machine learning models were evaluated using six performance metrics, including coefficient of correlation (R), root mean square error (RMSE), mean absolute error (MAE), relative absolute error (RAE), root relative squared error (RRSE), mean absolute percentage error (MAPE), and scatter index (SI), to ensure robust and reliable predictions. Additionally, SHapley Additive exPlanations (SHAP) analysis was conducted to perform a parametric identifying the influence of input parameters on Marshall Stability. Specifically, the XGB model achieved the best R values of 0.95 for training and 0.84 for testing, indicating strong correlations between predicted and actual MS values. Furthermore, SHAP analysis was conducted for the XGB model, which highlighted the significant influence of plastic size and bitumen content on MS prediction. SHAP and machine learning models can be used to optimize the composition of plastic-reinforced asphalt concrete for enhanced performance and sustainability. This research will provide practical guidance for pavement engineers and policymakers in utilizing waste plastic for sustainable infrastructure development.

1. Introduction

Plastic waste generation has become a major environmental challenge on a global scale. It is generally accepted that the harmful consequences of plastic waste are destroying life, making it an issue that affects everyone. According to estimates, the amount of plastic consumed has grown from 5 million tons during the 1950s to close to 100 million tons currently [1]. Around 360 million tons of plastic waste are produced across the world every year [2]. Plastic pollution has emerged as one of the current world's most urgent environmental problems [3]. Besides to current management techniques like recycling and energy recovery, new source-based solutions must be investigated for how to deal with the increasing amount of plastic waste [4]. There is a lot of focus on using waste materials throughout the pavement construction process without sacrificing the quality of the finished product [5]. One effective way to reduce the impact on the environment and decrease of natural resources is to recycle and utilize plastic waste [6].

A long-term structural analysis is necessary for traffic loads to be distributed at all types of road levels [7]. By maximizing material efficiency and minimizing maintenance requirements, Marshall Stability Mix Design ensures that asphalt pavements can tolerate high traffic volumes and environmental stressors while fostering sustainability, durability, and safety [8]. Pavement structures' behavior under various traffic loads and weather conditions can be predicted using the Marshall stability test. Asphalt concrete has a direct impact on the efficiency of infrastructure and road performance [9]. The resistance of asphalt concrete to settling, distortion, and displacement is assessed using Marshall stability. Over 110 million metric tons of asphalt are needed annually in the world, which means a substantial amount of money and energy are spent on preserving and revitalizing the world's current road infrastructure [10]. The utilization of waste plastic in the construction of roads has been deemed a sustainable paving method for its possible advantages for the environment and the economy [11]. Recycled aggregate concrete (RAC) highlights its material properties, mechanical

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behavior, and fire response, revealing that while extensive studies have examined the effects of recycled aggregates (RAs) on compressive strength and other mechanical properties, there remains a significant gap in understanding the high-temperature damage mechanisms and residual properties of RAC, particularly when incorporating fine RAs or a combination of coarse and fine RAs, which can influence spalling, failure modes, and overall structural safety under fire conditions [12].

Machine learning (ML) might take a great deal of physical labor and supplies to complete the traditional mix design process [13]. Better material design and performance prediction in civil engineering has been made feasible by machine learning, which not only increases predictive efficiency and accuracy but also provides a more nuanced understanding of the elements affecting asphalt binder performance [14]. ML has greater accuracy than conventional models and can handle big, complicated datasets while taking into account a variety of affecting factors [15]. Computerized optimization techniques based on (ML) and metaheuristic optimization algorithms provide an alternative to experimental mixture optimized for overcoming its limitations [16]. Therefore, it is essential to create computer vision-based models that automatically identify and measure the Marshall Stability of asphalt concrete reinforced with plastic. The field of machine learning (ML) is currently vital to research because of its capacity to solve numerous everyday issues involving statistics [17]. Artificial intelligence (AI) can find feature hierarchies and use them in various contexts. These algorithms automatically learn characteristics at different levels of abstraction, which allows them to model complex connections efficiently [18]. Predicting the Marshall Stability of plastic-reinforced asphalt concrete using metaheuristic machine learning algorithms, and subsequently using a multi-objective firefly approach to identify the ideal combination while taking strength into account [19]. Additionally, artificial intelligence techniques like Artificial Neural Networks (ANN) and Support Vector Machines (SVM) have emerged as effective tools for modeling the effects of aging on asphalt mixtures, demonstrating superior predictive capabilities compared to traditional methods [20].

Faramarzi et al. [21] explored the use of snail shell ash as a non-conventional filler in asphalt mixes to improve Marshall stability. Nwaobakata et al. [22] investigated the use of carbon nanotubes (CNTs) as an additive in asphalt mixtures to improve the technical characteristics of conventional asphalt pavements. Results showed that incorporating CNTs enhanced the Marshall stability of hot mix asphalt while reducing the Marshall flow, suggesting improved durability and performance. Ba-Nhan et al. [23] studied using the Extreme Gradient Boost (XGB) technique, which is optimized with Sailfish Optimizer (SFO) and Aquila Optimizer (AO), this work creates improved prediction models for Marshall Stability (MS) and Marshall Flow (MF) of basalt fiber asphalt concrete (BFAC). The results demonstrate that the XGB model, which has been cross-validated, has a high prediction accuracy. SFO is then used to optimize the BFAC composition design in accordance with current standards. Gul et al. [24] investigated Using a database of 343 data points, the study employed three soft computing approaches (ANN, ANFIS, and MEP) to construct prediction models for Marshall Stability (MS) and Marshall Flow (MF) of asphalt pavements. Model performance was assessed using a variety of statistical indicators. The MEP model performed better than the other approaches, offering MF and MS forecasts that were dependable, effective, and durable, which reduced on the expense and time involved in doing traditional laboratory testing. Kumar et al. [25] assessed how well soft computing models, such as artificial neural networks (ANN), Random Forest (RF), Random Tree (RT), Bagging RF, and Bagging RT, could predict the Marshall Stability (MS) of asphalt mixes made from plastic trash. Plastic size was found to be a significant factor influencing the MS of asphalt concrete including plastic waste by the sensitivity analysis, and the RF-based model demonstrated the highest accuracy and performance. Ghanizadeh et al. [26] studied the effectiveness of the Multivariate Adaptive Regression Spline (MARS) model, which outperforms traditional methods like Gene Expression Programming (GEP) in predicting flow number based on Marshall mix design parameters. Ghanizadeh et al. found EPR-TLBO model outperforms both the MGGP and existing empirical models, achieving a low interquartile range of absolute error (0.67 %). Additionally, the EPR-TLBO model demonstrated a high R^2 value of over 0.90 during both training and testing phases, indicating strong predictive accuracy in predict the air void (V) percentage of asphalt mixture [27]. Another studied focused on the use of artificial neural networks (ANN) to predict the rutting resistance and fatigue life of asphalt concrete and ANN models outperformed traditional regression models, offering a more accurate and robust prediction of asphalt mixture performance [28].

Arifuzzaman et al. [29] evaluated different machine learning (ML) models to predict important characteristics of plastic asphalt mixes (PAMs), like air vacancies, Marshall flow, and tensile strength. These models include decision tree (DT), AdaBoost regressor (AR), and bagging regressor (BR). The bagging regressor (BR) model was found to have the highest predictive accuracy, and the results indicate that ensemble techniques performed better than individual models when SHAP analysis was employed to evaluate the effect of input parameters on PAM characteristics. Salehi et al. [30] developed four ensemble machine learning (EML) models to predict the rheological properties of recycled plastic-modified bitumen (RPMB) based on experimental data, with CatBoost achieving the highest accuracy. Ankita et al. [31] studied that the stability of a carbon-fiber asphalt mix is evaluated using computational models. Based on a number of statistical metrics, ANN outperforms various models in this evaluation. Carbon fiber also contributes significantly to the stability of asphalt concrete, but bitumen content is shown to be the most relevant parameter, according to sensitivity analysis. Iftikhar et al. [32] With R^2 values of 0.89 for training and 0.88 for testing, the gene expression programming (GEP) model for estimating the compressive strength (CS) of plastic sand paver blocks (PSPB) is presented in this study with a high degree of accuracy. Sand size and fiber percentage were found to have a substantial impact on the CS through sensitivity analysis, indicating that the model is successful in encouraging sustainable construction and waste management practices.

This study promotes the use of waste plastic in asphalt concrete, addressing global challenges of plastic waste management while contributing to environmentally sustainable infrastructure development. In this research is to address a gap in knowledge and advance the field by examining the characteristics and possible applications of combinations of plastic waste. Machine learning methods offers a novel approach to understanding and predicting the mechanical properties of modified asphalt concrete, advancing the state-of-the-art in pavement material research. Furthermore, the study provides an in-depth feature importance analysis using SHAP. This research also incorporates sensitivity and uncertainty analyses, which have been largely overlooked in previous studies, ensuring a more robust and interpretable prediction framework. The findings contribute to the development of sustainable asphalt mixtures by optimizing plastic waste utilization while maintaining structural integrity.

This study aims to address this deficiency by: (1) To develop predictive models for the Marshall Stability (MS) of plastic-reinforced asphalt concrete using machine learning algorithms. (2) To evaluate and compare the performance of algorithms such as Random Forest (RF), Gradient Boosting, Extreme Gradient Boosting (XGB), and Bagging Regression (BR) in predicting MS. (3) To identify and analyze the key factors influencing MS using SHapley Additive exPlanations (SHAP). (4) To conduct uncertainty analysis to quantify and understand the reliability and confidence of the predictions made by the machine learning models. The outcomes can guide engineers and policymakers in designing durable and sustainable roadways, reducing maintenance costs and enhancing road performance under diverse conditions.

Table 1
Physical characteristics of aggregates.

Index (Unit)	Fine aggregate	Coarse aggregate	Standard
Water absorption (%)	2.1	2.89	ASTM C-128
Bulk density (g/cm^3)		1.45	
Bulk specific gravity	1.74	3.1	
Apparent specific gravity	2.61	2.91	
Crushing value of aggregate (%)		17.13	ASTM C-127
Impact value of aggregate (%)		11.34	
Aggregate abrasion value (%)		29.3	ASTM C-131
Flakiness index (%)		15.3	ASTM D-4791
Elongation index (%)		9.9	
Specific gravity	2.13		ASTM D-792
Fineness (%)	1.99		

Table 2
Bitumen characteristics.

Index	Value	Standard
Flash point test	277^0 C	ASTM D-92
Penetration test	96.53 mm	ASTM D-5
Ring and ball test	41.3^0 C	ASTM D-36
Specific gravity	1.02	ASTM D-70

2. Materials and methodology

In this research, materials were selected in the literature review investigation, such as asphalt graded as viscosity grade (VG)-30, marble powder as a filler, and coarse aggregate with a nominal size of 20 mm [33–37]. The characteristics of these materials are as follows:

Aggregate:

The asphalt mixture was formulated using a nominal coarse aggregate size of 20 mm. Table 1 lists the coarse and fine aggregates' physical characteristics. To give the asphalt mixture the right consistency, marble powder which accounted for around 10 % of the weight of the coarse aggregate was added as a filler.

Bitumen:

The bitumen utilized in this study was VG-30. The penetration grade of this specific bitumen type ranges from 60 to 70. The primary components of asphalt are described in Table 2.

Waste plastic:

Three different plastic types like PET, HDPE, and PVC—are combined in different proportions in this study. This makes it possible to thoroughly examine how they affect the goals of the study. The study learns more about how various ratios work together to influence desired results.

2.1. Data collection

The dataset used in this study consists of a total of 265 observations compiled from existing literature [8,9,23–25,33,35,37–39]. The inputs parameters influencing the Marshall Stability (MS) of plastic-reinforced asphalt concrete, such as bitumen content, Plastic Content, aggregate, and plastic size. Table 3 summarizes the descriptive statistics of the dataset used in this study. MS ranges from 0.05 to 34.23 kN, with a mean of 15.67 kN. These statistics provide a comprehensive view of the dataset's diversity and range. Fig. 1 displays the multi-correlation matrix (Heatmap) of the input and output points for the investigation. The various correlation values are shown by the different colors. To explore the correlation coefficient using the Pearson correlation coefficient method in order to investigate the relationship between the variables in the data set. Plastic content exhibits the highest positive correlation with MS ($r = 0.22$), suggesting that an increase in plastic content can moderately enhance the stability of the asphalt concrete mix. But bitumen grade shows a negative correlation with MS ($r = -0.23$). Plastic Size has strong correlations with other input features but not directly with MS. Bitumen content and aggregate demonstrate weak positive correlations with MS ($r = 0.06$ and $r = 0.09$, respectively), indicating a relatively minor influence on Marshall Stability. Thus, plastic content plays a more significant role in enhancing Marshall Stability than the other parameters.

2.2. Machine learning models

The proposed method machine learning (ML) models to predict the Marshall Stability of plastic-reinforced asphalt concrete. To assess the models' performance and generalization capabilities, the dataset was split into 70 % for training and 30 % for testing. The selected ML models were then trained on the dataset, and their accuracy was evaluated using multiple performance metrics to ensure robust and reliable predictions.

2.2.1. Random forest

Random forest method is an appreciated collective learning approach for regression and classification. So as to increase accuracy and resilience, it builds several decision trees during training and then aggregates their results [40]. Leo Breiman developed the Random Forest (RF) algorithm, a machine learning technique that builds a "forest" of decision trees to process multidimensional, complicated data. Using random sampling with replacement, the RF algorithm generates several samples from the dataset, make sure that each sample is marginally unique [41]. A separate decision tree is trained using each sample subset. RF chooses a subset of features (variables) to take into account at random for every decision tree split. This reduces overfitting and enhances the model's capacity to handle high-dimensional data by limiting each tree's dependence on particular variables. In categorization, RF aggregates each tree's output by deciding on a final class label by a majority vote, but in regression, it averages all of the trees' predictions [42,43]. By monitoring each feature's impact on model correctness across trees, RF offers metrics of variable importance. In bioinformatics, where certain variables may be more biologically relevant, it is especially helpful in determining which variables have the greatest influence on outcome prediction. RF is incredibly useful in bioinformatics due to its resilience to complicated, unbalanced datasets and its capacity to handle significant correlations between variables, a common occurrence in proteomic and genomic research [44].

Table 3
Descriptive statistics of the dataset.

Statistic	Bitumen Content (%)	Plastic Content (%)	Bitumen Grade	Aggregate (mm)	Plastic Size (mm)	Marshall Stability (kN)
Mean	5.356	5.340	25.66	18.83	7.15	15.67
Std Dev	0.778	3.785	7.812	2.821	4.142	4.315
Min	4.000	0.000	10.00	9.500	2.360	0.050
25 %	5.000	3.000	20.00	20.00	3.000	12.80
50 %	5.500	6.000	30.00	20.00	4.750	15.49
75 %	6.000	9.000	30.00	20.00	12.00	18.40
Max	10.000	20.000	30.00	20.00	12.00	34.23

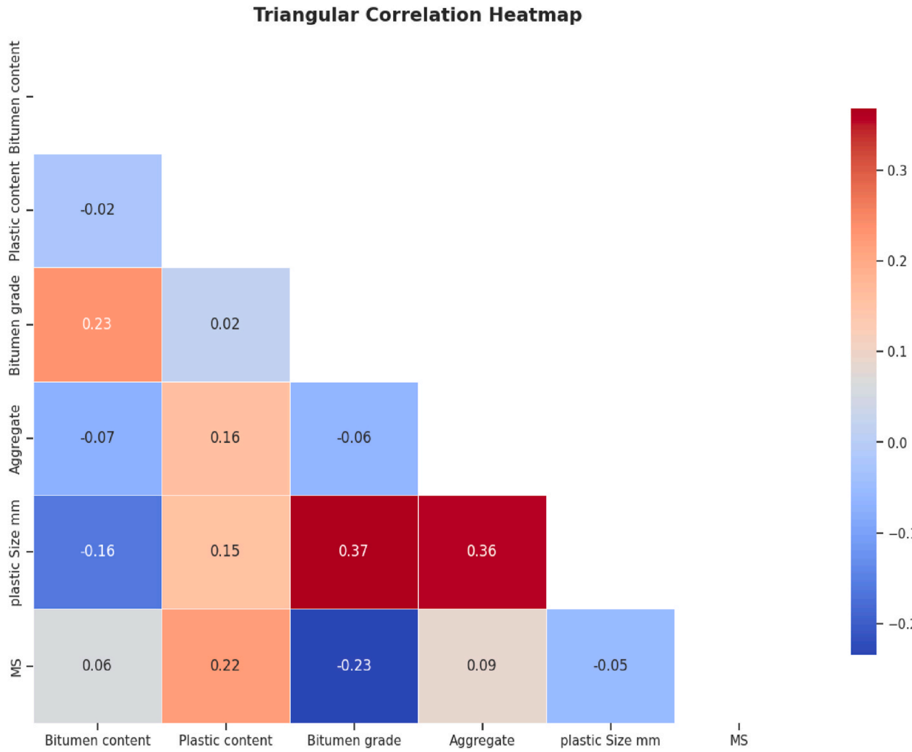

Fig. 1. Heatmap of correlation between input and output variables.

$$Y = \frac{1}{T}\sum_{t=1}^{T} Y_t \qquad \text{Eq. 1}$$

Y: Final predicted value (average prediction).
T: Total number of decision trees in the forest.
Y_t: Prediction from the t-th decision tree.

2.2.2. *Gradient boosting regression*

A machine learning method called gradient boosting is applied to classification and regression problems. In order to increase prediction accuracy, it merges several weak models usually decision trees into a stronger ensemble [43,45]. Each new model in the iterative process seeks to fix the mistakes of the one before it. This is accomplished by paying close attention to the gradient, or direction of steepest descent, of the loss function, which quantifies the prediction error of the model [46]. The algorithm determines the loss gradient at each stage and applies it to modify weights, giving priority to fixing large errors. The model's accuracy is increased during iterations as a result of this "boosting" technique. For complicated datasets, gradient boosting works especially well because it balances variance and model bias [45].

2.2.3. *Extreme Gradient Boosting*

A well-known machine learning technique called the XGBoost model works on the boosting principle, which combines several weak learners (usually decision trees) to produce a powerful prediction model [47]. A model is initialized at the start of the process, and trees that fix the mistakes of the earlier ones are added iteratively. With an emphasis on the cases that were incorrectly classified, each tree is constructed using the residuals of the predictions. Gain, Cover, and Frequency are some of the metrics that XGBoost offers in relation to feature relevance. Gain quantifies the contribution a feature makes to the model, Cover shows the proportion of observations a feature affects, and Frequency counts the number of times a feature appears in the trees [43]. This improves interpretability and directs additional research or feature selection in landslide susceptibility mapping by enabling users to determine which features have the most influence on result prediction [43].

Here's the basic formula for predicting a value Yi for an instance i:

$$Y_i = \sum_{k=1}^{K} f_k(x_i) \qquad \text{Eq. 2}$$

where:
Y_i is the predicted output for instance i1.
X_i the feature vector for instance i1.
K is the total number of trees,
f_k is the function (or "weak learner") represented by the k-th decision tree in the model.

2.2.4. *Bagging regressor*

The Bagging Regressor is a useful machine learning method for handling complicated data and guaranteeing prediction consistency in mix design. It lowers variance and improves model robustness by averaging the predictions of several models trained on various data subsets [48]. Even when working with noisy or high-dimensional datasets, our ensemble technique reduces the chance of overfitting and guarantees consistent and dependable results. In resource-intensive applications where accuracy and stability are essential, the Bagging Regressor performs exceptionally well. It is the best option for verifying mix design predictions since it offers a reliable standard for baseline performance. Its capacity to combine many model outputs produces more precise and balanced results while guaranteeing resistance to data fluctuations and outliers [49]. The Bagging Regressor is a crucial instrument for enhancing the accuracy and dependability of forecasts in demanding computational and engineering applications because of its stability and efficacy [50].

$$Y = \frac{1}{M}\sum_{m=1}^{M} Ym \qquad \text{Eq. 3}$$

Where,
Y: The final predicted value of the Bagging Regressor.
M: The total number of base models in the ensemble.

Table 4
Model hyperparameters for machine learning models.

Model	Hyperparameter	Range	Optimal Value
Random Forest	learning_rate	0.01, 0.1, 0.2, 0.5	0.1
	max_depth	3,5,9,11,20	11
	n_estimators	50,80,100,200,300	200
	min_samples_leaf	1, 2, 4	1
	min_samples_split	2,5,10	2
Gradient Boosting	n_estimators	100,200,300,500	200
	learning_rate	0.01, 0.1, 0.2	0.1
	max_depth	3,5,7,9,11	5
	subsample	0.1, 0.5,0.8,1,2	0.8
XGBoost	learning_rate	0.01, 0.1, 0.2, 05	0.1
	max_depth	3,5,7,9	5
	n_estimators	100,200,300,500	200
	subsample	0.1, 0.5,0.8,1,2	0.8
	colsample_bytree	0.7, 0.8, 0.9	0.9
Bagging Regressor (BR)	n_estimators	100,200,300,500	100
	max_samples	0.1, 0.2, 0.5, 0.7,1	0.7
	max_features	0.1, 0.2, 0.5, 0.8,1	1
	bootstrap	True, False	True
	bootstrap_features	True, False	True

Ym: The prediction from the m-th base model.

2.3. Hyper parameter tuning with grid search

Grid search is a hyperparameter optimization technique used to systematically explore a predefined set of hyperparameter values for a machine learning model. Grid search is a robust and exhaustive hyperparameter tuning method, particularly effective for small to medium-sized search spaces [17,30,31]. Table 4 presents the hyperparameter tuning results for four machine learning models: Random Forest, Gradient Boosting, XGBoost, and Bagging Regressor. Optimal values were determined using grid search to enhance predictive accuracy. Key parameters such as learning rate, max depth, n_estimators, subsample, and feature selection were optimized. Random Forest performed best with max_depth = 11 and n_estimators = 200, while XGBoost and Gradient Boosting favored max_depth = 5 and n_estimators = 200. Bagging Regressor achieved optimal performance with max_samples = 0.7 and bootstrap = True. These optimized settings ensure robust model performance while preventing overfitting and computational inefficiency.

2.4. SHapley additive exPlanations (SHAP)

Shapley values, a mathematical method for calculating each participant's contribution in a cooperative game, are used in SHAP, which is based on game theory. When a machine learning algorithm is used, these players are the input variables, and the game represents the expected results [51]. By assessing how the model's predictions vary when a feature is added or removed from the input, the impact of each feature in a model is evaluated. It provides insights into how individual features contribute to the predictions made by a model [52]. The Shapley value ϕ_i for feature i is determined by calculating the average marginal contribution of that feature across all possible permutations of features. In this equation, N represents the set of all features, S represents a subset of features that excludes feature i, $|S|$ denotes the cardinality of set (S), $v(S)$ represents the model's prediction when only features in set (S) are considered, and $v(S\cup\{i\})$ represents the model's prediction when feature i is added to set S.

$$\phi_i = \sum_{S\subsetneq N\{i\}} \frac{|s|!(|N|-|S|-1)!}{|N|!}[v(S\cup\{i\})-v(S)] \quad \text{Eq. 4}$$

2.5. Model performance parameters

In this study thoroughly analyzed the predictive performance of RF, GBM, XGB, and BR models in forecasting the MS of waste plastic-modified asphalt concrete. Six performance metrics—CC, MSE, RMSE, MAPE, RAE, and SI were used to evaluate the ML models prediction performance. This comprehensive evaluation aimed to assess the effectiveness of each model in MS prediction, highlighting their strengths and weaknesses, and offering valuable insights for improving decision-making in asphalt engineering. The calculations for these metrics are provided in Equations 5–10.

$$R = \left(\frac{\sum_{d=1}^{D}(m_d-\overline{m})(z_d-\overline{z})}{\sqrt{\left[\sum_{d=1}^{D}(m_P-m)^2\right]\left[\sum_{d=1}^{D}(z_d-\overline{z})^2\right]}} \right) \quad \text{Eq. 5}$$

$$RMSE = \sqrt{\frac{1}{D}\sum_{d=1}^{D}(z_d-m_d)^2} \quad \text{Eq. 6}$$

$$MSE = \frac{1}{D}\sum_{d=1}^{D}(z_d-m_d)^2 \quad \text{Eq. 7}$$

$$\text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{M_i-F_i}{M_i}\right| \quad \text{Eq. 8}$$

$$RAE = \frac{\sum_{d=1}^{D}|m_d-z_d|}{\sum_{d=1}^{D}|m_d-\overline{m}|} \quad \text{Eq. 9}$$

$$\text{SI} = \frac{RMSE}{\overline{x_i}} \quad \text{Eq. 10}$$

Where.

m_d	Predicted value at instance d
z_d	Actual (observed) value at instance d
$\overline{m}$	Mean of predicted values
$\overline{z}$	Mean of actual values
D	Total number of data points
M_i	Actual value for instance i
F_i	Predicted value for instance i
n	Total number of instances in MAPE calculation
$\overline{x_i}$	Mean of observed values used in Scatter Index calculation

3. Results and discussion

3.1. Results of ML models

The line graph of the actual and predicted values for both stages, including error, is shown in Fig. 2. As shown in Fig. 2(a) BR model demonstrates the most of the predicted data points are approximately close to actual data at the training phase and the error value ranges from −5 to +5. In Fig. 2(b), the testing phase shows most of the predicted data points moderately adjusted with actual data points. In Fig. 2(c), the maximum points of actual and predicted data points in the GBM model was well matched, but the minimum actual data points could not be predicted well, and the error value showed −10 to +8. Shown in Fig. 2 (d), the maximum predicted datasets of the GBM model were well-adjusted with the actual datasets and the the error value showed −6 to +5 at the testing stage. In Fig. 2(e), at the training stage, the RF model, the maximum data points was well predicted, but the minimum value of actual data could not be predicted well, and the error value ranged from −10 to +5. In Fig. 2(f), the RF model moderately adjusted

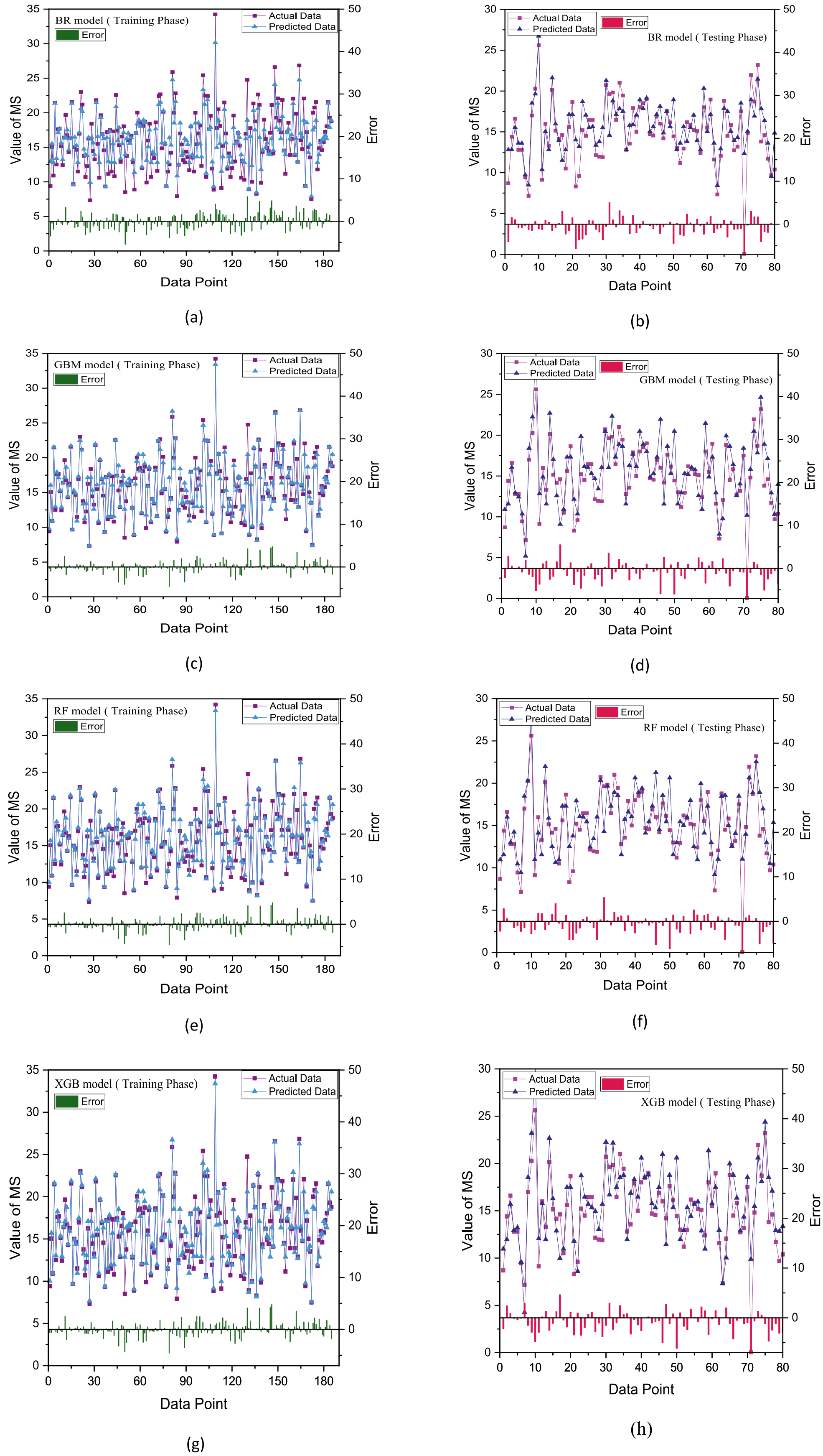

Fig: 2. Observed and Predicted values' distribution for machine learning models.

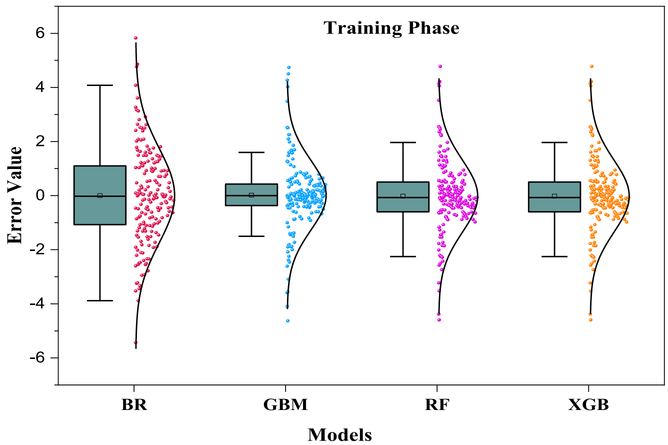

Fig: 3. Error distribution analysis of Marshall stability prediction at training stage.

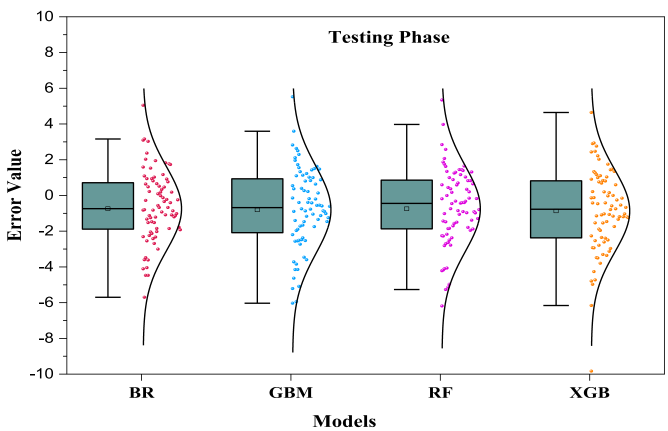

Fig: 4. Error distribution of Marshall stability ML prediction at testing stage.

between actual and predicted data points and the error value range was −5 to +5. As shown in Fig. 2(g), the XGB model performed well because the largest number of the predicted data points were very close to the real data points and the error value range of −3 to +3. At the testing stage, in Fig. 2(h), the XGB model represented the lower error range, where the error value from −5 to +5. For both MS property phases, all machine learning models' assessed errors were less than 10 kN. Plotting of the observed and predicted data points is done jointly. Both phases are tightly and well matched by the actual and predicted values. However, compared to other models in both phases, the XGBoost model performs the best, showing a stronger connection between actual and predicted values.

Figs. 3 and 4 illustrates the distribution of error values for Marshall Stability prediction using four machine learning models. The violin plots combined with boxplots provide a comparative visualization of the error spread and central tendency during both the training (Fig. 3) and testing (Fig. 4) phases. In the training phase, all models exhibit relatively symmetrical error distributions centered around zero. Among them, GBM and RF models show the narrowest interquartile ranges and lowest variance, suggesting higher prediction stability and less overfitting. In contrast, BR and XGB models exhibit slightly wider spreads, with BR model displaying a few more outliers. During the testing phase, the spread of error increases slightly for all models, which is typical due to exposure to unseen data. GBM and RF model continue to maintain compact distributions, demonstrating good generalization capability. However, BR and XGB models show comparatively higher variance and a broader range of errors. Overall, GBM and RF models showed to perform better in terms of prediction accuracy and stability for Marshall Stability values in both phases, as reflected by their tighter boxplots and lower spread in error distributions.

Fig. 5 shows the relationship between the predicted and actual values of Marshall Stability (MS) for four machine learning models. The scatter plots represent the performance of each model during both the training and testing phases. Most of the machine learning-predicted MS data points (RF, GBM, XGB, and BR) had an error range between −15% and +15%, but a small number of data points had errors that were away from that range. Across all models, the testing phase tends to show a slightly larger spread in the data points, indicating the models' slight decline in predictive accuracy when applied to new, unseen data. However, all models remain within reasonable error margins, with the prediction errors staying within ±15 %.

3.2. Comparison of the results ML models

Fig. 6 shows the performance of four machine learning models. The R-value of four algorithms is shown in Fig. 6. For all algorithms, the training set's R-value is always greater than the testing set's. In the training phase, the model that uses BR has the lowest R score of all the models; XGB, RF, and GBM all have the same value, which is 0.956. The model utilizing BR had the greatest values of all five models in terms of MSE, RMSE, MAPE, RAE, and SI (Fig. 6) This indicates that BR has the lowest accuracy, and the findings of the R score were in good agreement with each other. For the testing part, the XGB model has the lowest MSE, RMSE, MAPE, and SI values, and its R value is higher than that of any other model. For the same part, the GB model has the highest MSE, RMSE, RAE, and SI values. Extreme Gradient Boosting (XGB) performs the best overall, with low MSE, RMSE, MAPE, and SI values and a high R score. The comparison of the Marshall stability prediction performance of four machine learning algorithms, as shown in Table 5, evaluates their prediction abilities using various performance metrics (R, MSE, RMSE, MAPE, RAE, and SI). A higher R-value and smaller values of MSE, RMSE, MAPE, RAE, and SI indicate better model performance. For the training set, all algorithms performed well, with R-values above 0.950, except for the BR algorithm. In the testing set, all algorithms had high R-values (above 0.820), except for BR and RF. Among the algorithms, XGB demonstrated the highest prediction ability, while BR had the lowest.

3.3. Uncertainty analysis

Uncertainty analysis involves evaluating the reliability and accuracy of machine learning model predictions. This methods may result in predictions that are overly optimistic because they make use of big data sets. However, Fig. 7 demonstrates the results of this research, which assessed the uncertainty value of four models used. For the testing phase, in Fig. 7 (a), it is observed that the XGBoost model slightly better performance than other models such as RF, BR, and GBM. The XGBoost model provided a lower uncertainty value, which is around 6.518, and the lowest value among other models. The uncertainty value was found to be highest from the gradient boosting model. For the training stage, the XGBoost, RF, and Gradient Boosting models found the uncertainty value lower than the BR model in Fig. 7 (b). But overall, all models display minor uncertainty values, which indicate good performance. GBM and XGBoost models displayed a lower uncertainty value, which is around 3.6, and the lowest value among other models.

3.4. Sensitivity analysis with SHAP

The SHAP analysis conducted in this study provides a detailed understanding of the influence of input variables on the hybrid model's predictions. Fig. 8 illustrates the mean SHAP values, computed by averaging individual SHAP values across the entire dataset, which highlight the importance of features in determining the model's output. Features with larger absolute mean SHAP values are deemed more

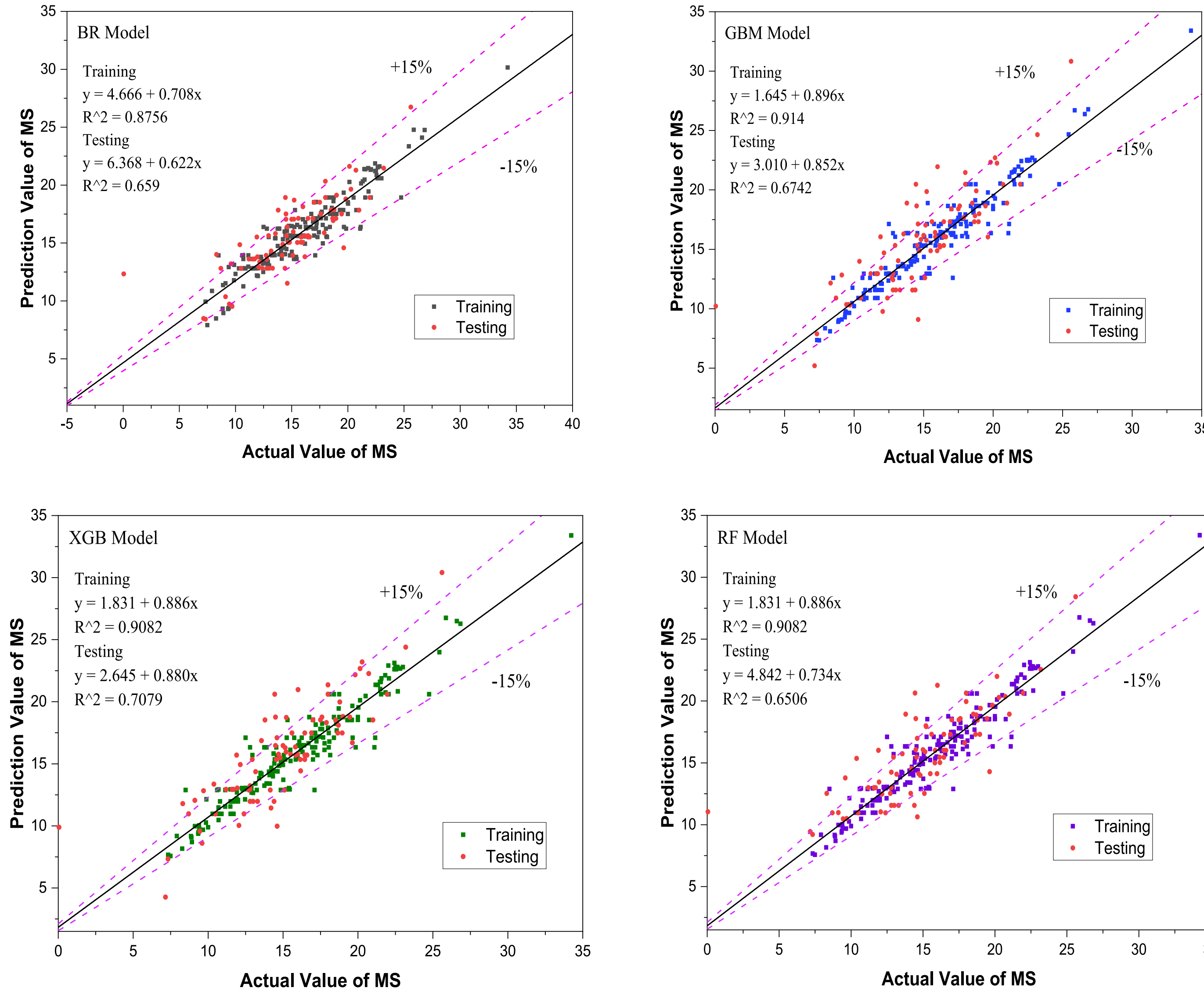

Fig: 5. Relationship between predicted and actual values.

influential in predicting Marshall Stability (MS).

According to Fig. 8, plastic size emerges as the most significant feature, with a mean SHAP value exceeding 1.2. This indicates that variations in plastic size have a substantial impact on MS predictions. Larger plastic sizes contribute positively to the strength and stability of asphalt concrete, thereby enhancing MS. Bitumen content, functioning as a binder, ranks as the second most influential factor, demonstrating a marked effect on MS. An increase in bitumen content improves the cohesiveness and binding properties of the material, directly enhancing its stability. Plastic content follows as the third most impactful variable, further emphasizing its critical role in improving MS. Additional influential factors include bitumen grade and aggregate size. Bitumen grade strengthens the material by improving durability and cohesion, while aggregate size influences the structural stability and integrity of the mixture.

To further analyze global feature relevance, SHAP values were used to rank the variables in descending order of importance. Fig. 9 displays the relative importance of each feature using a scatter plot with colored dots, where the x-axis represents the SHAP level and the y-axis indicates feature relevance. The gradient of colors in the plot ranging from blue (lower importance) to red (higher importance) provides a clear visual representation of how each input variable contributes to the model's predictions. This plot also reveals the direction and magnitude of each feature's impact on the MS. The results of the SHAP analysis not only highlight the dominant role of plastic size and bitumen content in MS prediction but also provide actionable insights for optimizing asphalt concrete mixtures. These findings emphasize the utility of SHAP as a powerful interpretability tool for machine learning models in civil engineering applications.

4. Conclusion

The current study comprehensively evaluated the performance of machine learning models of plastic-reinforced asphalt concrete. The findings of this study are summarized as follows:

- Among the models, XGB demonstrated superior performance, as evidenced by its higher correlation coefficient (CC) and lower error metrics, including RMSE, MSE, and MAPE, showcasing its robustness performance.
- The SHAP analysis for the XGB model provided valuable insights into the significance of key input parameters. It revealed that plastic size and bitumen content had the most significant influence on MS prediction.

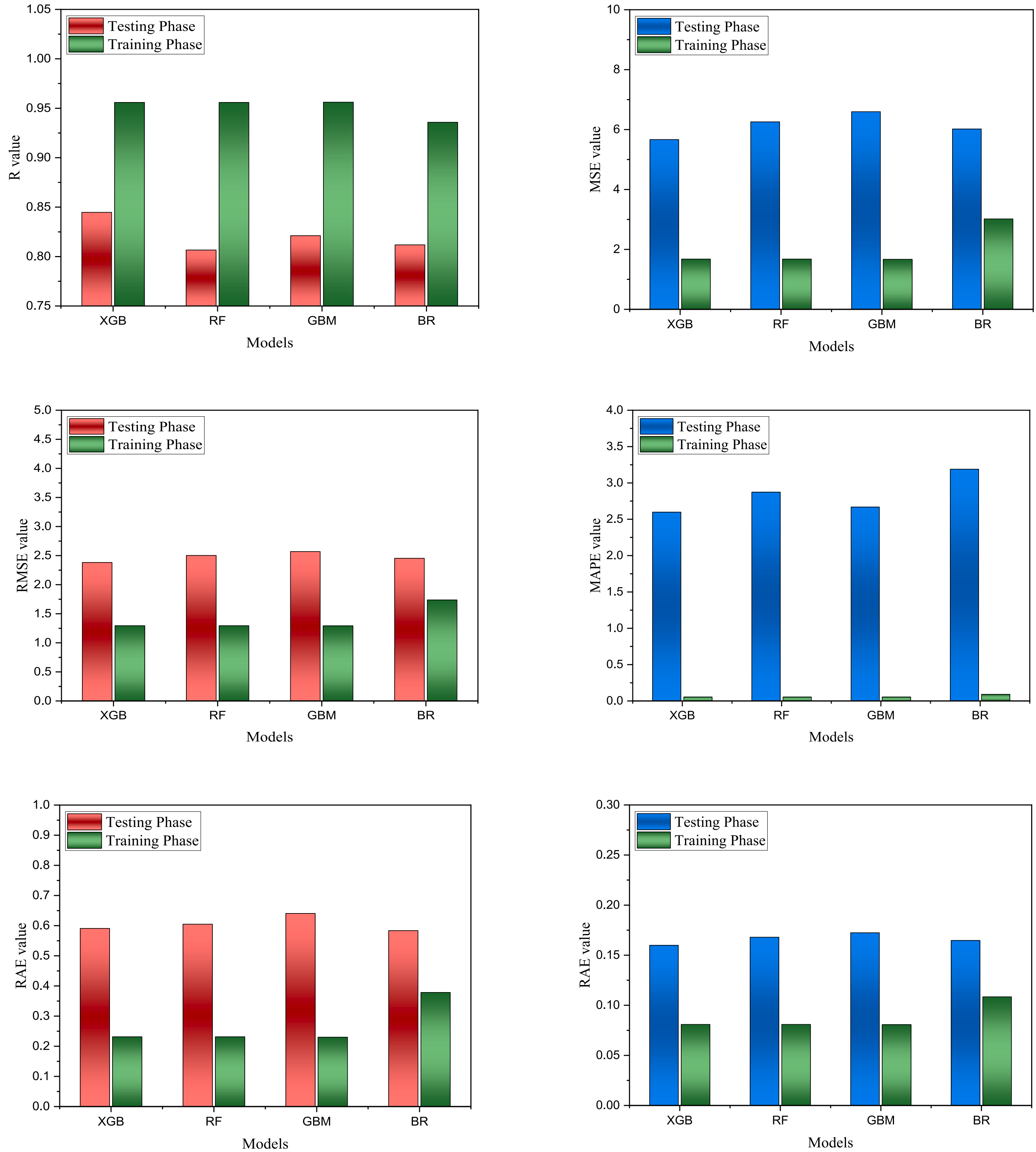

Fig: 6. Performance compare of machine learning models.

- GB and XGB exhibited lower uncertainty, indicating more reliable predictions.
- The findings of this research highlight the potential of machine learning models, particularly XGB, as powerful tools for enhancing the design and quality of sustainable pavement materials.

Machine learning models can contribute to more efficient decision-making in asphalt engineering, paving the way for advancements in material performance and environmental sustainability. The study depends on dataset obtained from previously published works. Asphalt concrete gradation and plastic type were not included, which could significantly affect the model's predictive accuracy for the Marshall Stability of plastic-reinforced asphalt concrete.

Future studies can focus on incorporating larger and more diverse

Table: 5
Model performance parameters.

		XGB	RF	GBR	BR
Training Metrics	CC	0.956	0.956	0.956	0.936
	MSE	1.675	1.675	1.668	3.016
	RMSE	1.294	1.294	1.291	1.737
	MAPE	0.053	0.053	0.053	0.09
	RAE	0.231	0.231	0.229	0.378
	SI	0.081	0.08	0.08	0.108
Testing Metrics	CC	0.845	0.807	0.821	0.812
	MSE	5.665	6.256	6.596	6.015
	RMSE	2.38	2.501	2.568	2.453
	MAPE	2.598	2.872	2.667	3.189
	RAE	0.59	0.605	0.64	0.583
	SI	0.159	0.168	0.172	0.165

datasets, exploring hybrid modeling approaches, and applying these methodologies to other pavement-related parameters. Future research considers additional input variables such as asphalt gradation, plastic types, and environmental factors, could improve model performance. Advanced methods for hyperparameter tuning, such as Bayesian optimization or metaheuristic algorithms, can further reduce overfitting and improve model generalization.

CRediT authorship contribution statement

Mahmudul Haque Jamil: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Ravi Jagirdar:** Writing – review & editing, Software, Methodology, Investigation, Conceptualization. **Abul Kashem:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **MD Nimar Ali:** Writing – review & editing, Resources, Investigation, Formal analysis, Conceptualization, Data curation. **Dipongkar Deb:** Writing – original draft, Data curation.

Ethics approval

No participation of humans takes place in this implementation process.

Consent to participate

Not applicable.

Consent to publish

All authors mutually agreed to publish the work in this journal.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

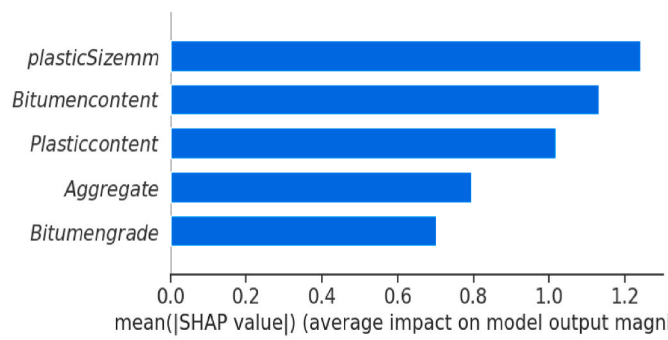

Fig. 8. A sensitivity study of SHAP for the model that performed the best model.

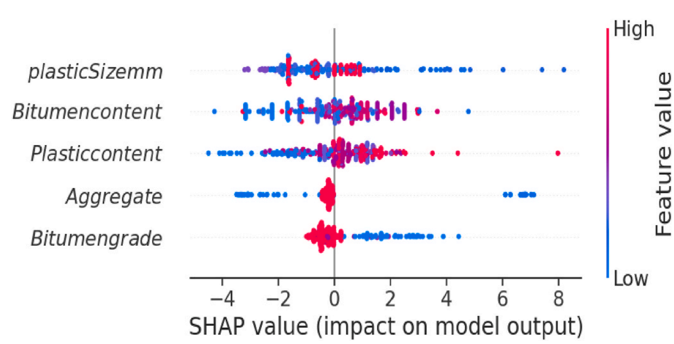

Fig. 9. Summary plot of SHAP values.

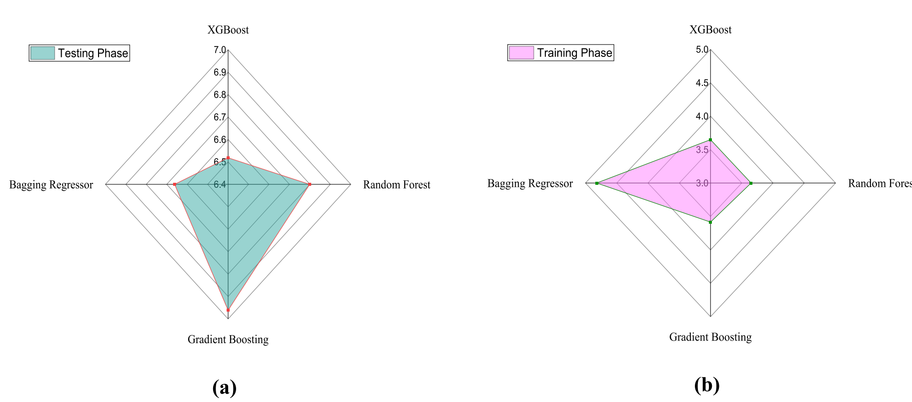

Fig. 7. ML models are evaluated using the uncertainty test.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.hybadv.2025.100483.

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